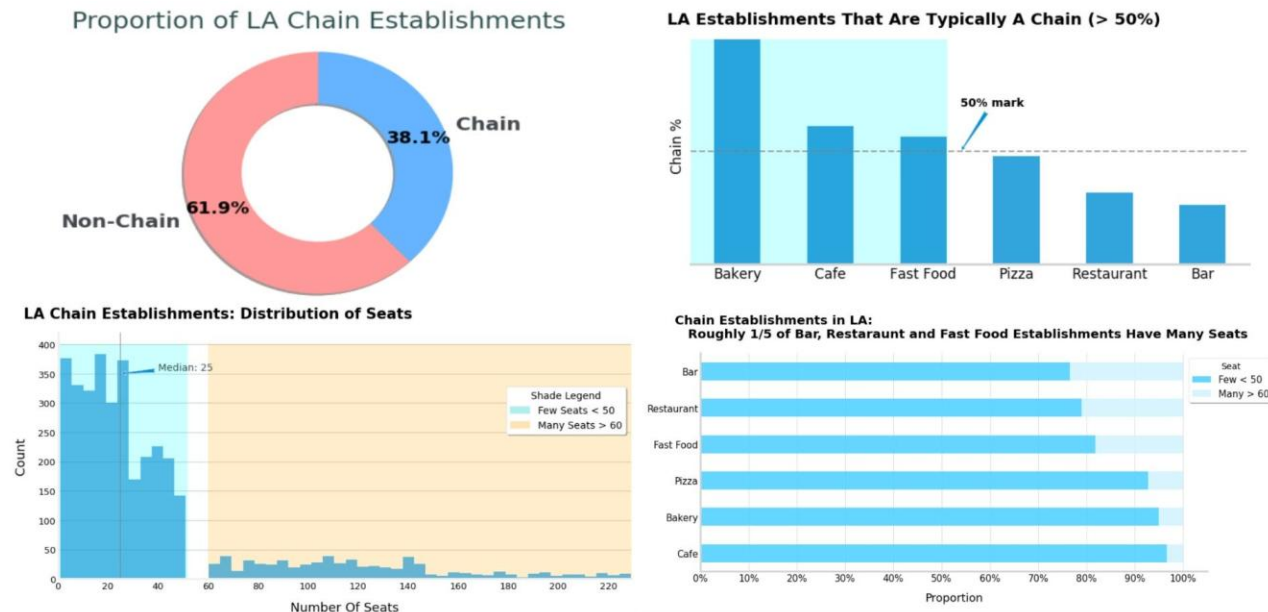

THE OBJECT-ORIENTED (OO) PARADIGM



- **The Concept:** Moving beyond `plt.plot()` (Functional) to the `fig, ax` structure (Object-Oriented).
- **The "Canvas" Analogy:**
 - The **Figure** is the physical canvas or window.
 - The **Axes** are the individual charts or "subplots" sitting on that canvas.
- **Why it matters:** This approach provides granular control over every pixel, label, and tick mark, which is essential for multi-panel professional reports.

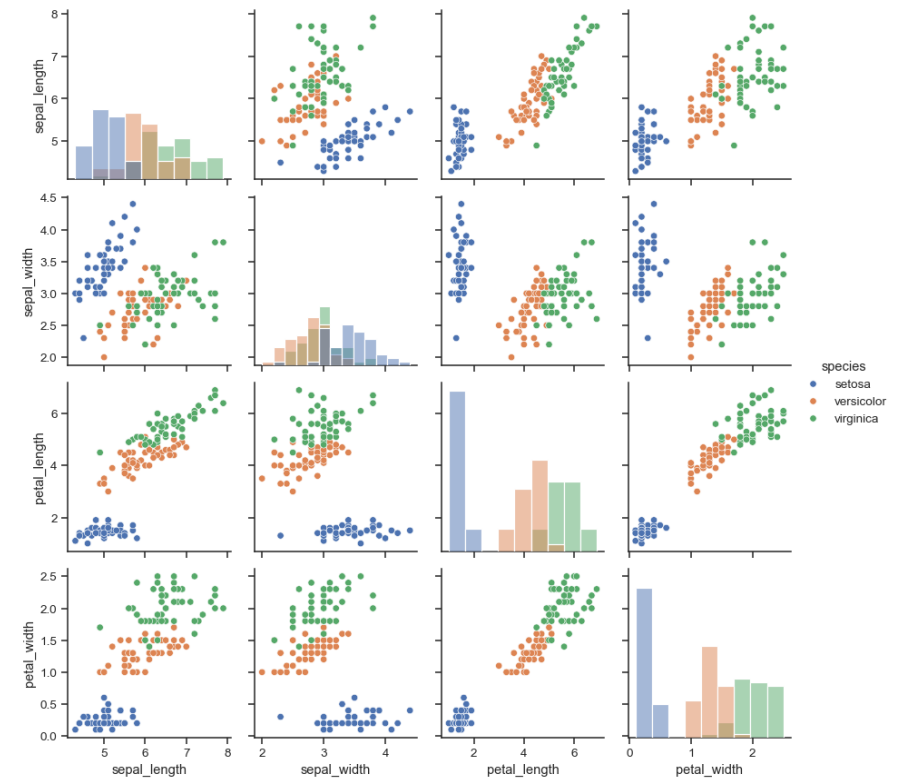
STATISTICAL STORYTELLING



- **Beyond Bar Charts:** Simple averages often hide the truth.
- **Box-plots:** Visualizing the "Five-Number Summary" (Min, Q1, Median, Q3, Max) and detecting outliers.
- **Violin-plots:** Combining box-plots with a density curve to show the "shape" and "frequency" of data.
- **Joint-plots:** Seeing the relationship between two variables and their individual distributions at the same time.

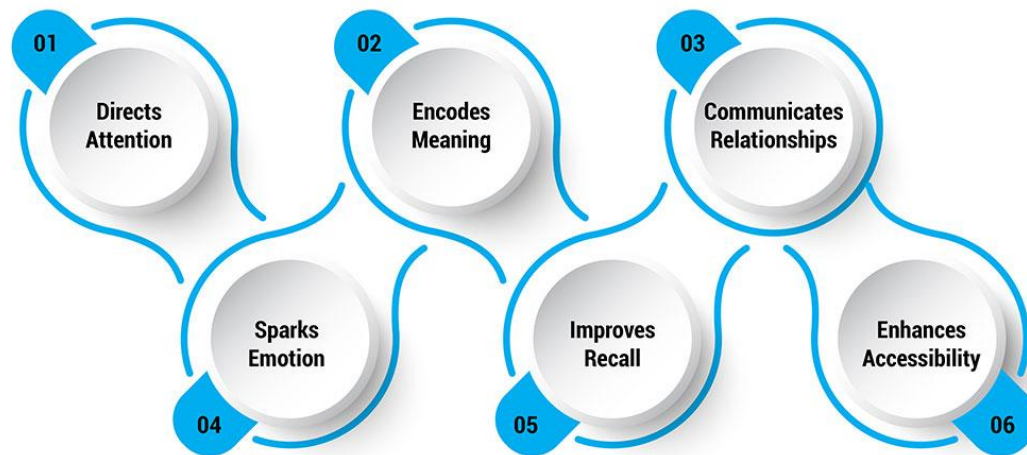
FACETGRIDS (SMALL MULTIPLES)

- **The Dimensionality Problem:** How do you show a trend across 10 different cities or 5 different years in one view?
- **FacetGrids:** Breaking a single dataset into a grid of multiple small charts based on a category.
- **Efficiency:** It allows for immediate visual comparison across subsets of data without cluttering a single chart.



COLOR THEORY & ETHICS

The Role of Color in Data Visualization



- **The Narrative of Color:** Colors should highlight data, not distract from it.
 - **Professional Palettes:** Moving away from "Default" colors to "Sequential" (for levels), "Diverging" (for positive/negative), and "Qualitative" (for categories).
 - **Accessibility:** Ensuring charts are **Colorblind-friendly** (using palettes like 'viridis' or 'colorblind') to ensure information is accessible to all stakeholders.
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